

# ARYAN MAHESHWARI

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## EXPERIENCE

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### Founder & Lead Engineer

May 2025 – Present

#### **Aero Robotics (Student Venture)**

- Built a drone-deployed robotic system for hazardous industrial environments, starting with wind turbine blade inspection as the first commercial application; project is in active development
- Designed and built a sensor payload for wind turbine blade maintenance covering surface prep, composite-safe cleaning, adhesive application, and precision sensor placement within payload weight limits; USPTO patent application filed (Sept. 2025)
- Designed and prototyped a universal drone attachment platform with modular quick-swap payloads; iterated 3D-printed prototypes in Fusion 360 for vibration isolation, structural stability, and cross-platform compatibility
- Piloted 20 field deployments at 125m for a commercial wind energy client in Minnesota as an FAA-certified UAS operator; pursuing Part 107 commercial certification

### Commercial Operations Intern

Jun 2024 – Aug 2024

#### **Mars Materials (ClimateTech, backed by Breakthrough Energy)**

- Worked directly with the CEO on fundraising, preparing investment decks, reviewing investor theses, and drafting outreach communications
- Visited the Mars Materials pilot system at Shell Technology Center, documented post-run technical debriefs, and turned them into reusable operational workflows
- Contributed to grant applications and internal employee recognition program design

### Product Design Intern

May 2023 – Aug 2023

#### **Predyct (ClimateTech)** · predyct.io

- Researched wind turbine blade failure statistics and low-power sensing requirements; findings fed directly into the company's core sensor specification
- Ran competitive analysis on blade monitoring solutions to map market gaps that shaped Predyct's product direction

## EDUCATION

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### University of Illinois Urbana-Champaign

Aug 2026 – May 2030

B.S. Aerospace Engineering · Minor in Business · Expected May 2030

### Lawrence E. Elkins High School, Engineering Academy

2022 – 2026

Honors Diploma · Specialization in Engineering & Computer Science · Fort Bend ISD

## TECHNICAL SKILLS

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**Flight Systems:** ArduPilot, Pixhawk-class flight controllers, GNSS/RTK integration, ESC and radio systems, battery and payload budgeting; FAA Recreational Flyer Certificate

**Mechanical Design:** Autodesk Fusion 360 (Certified), 3D prototyping, payload structural design, mounting systems

**Programming:** Python, Java, HTML/CSS

**Electronics:** Soldering, circuit integration, sensor systems, connector and wiring selection

**Other:** AI/ML fundamentals, investment deck preparation

## HONORS & ACTIVITIES

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- Student Body Historian, Elkins High School (2025–2026)
- NASA High School Aerospace Scholars (HAS) Program (Oct 2024 – Jul 2025)
- TSA Drone Challenge, Pilot, State Finalist (2026)
- TEX-E ClimateTech Entrepreneurship Bootcamp (Apr 2025)
- Academic Excellence Award, Elkins High School (2022–2025)
- District Award, VEX Robotics Competition (2023)
- Varsity Golf, 4 Athletic Awards (2022–2025)
- 1st Degree Black Belt, Tae Kwon Do (2023–2025)
- Student Teacher & Cultural Volunteer, Chinmaya Mission Houston (2018–Present)